Solving systems of equations with a v[redacted]ty of methods

1 The point of intersection of two lines.

2 -3

3 -4

4

a

x	$y=3x-1$	$y=-5x+23$
1	2	18
2	5	13
3	8	8
4	11	3
5	14	-2

b $x=3, y=8$

5

a

x	$y=5x$	$y=9-4x$
1	5	5
3	15	-3
5	25	-11

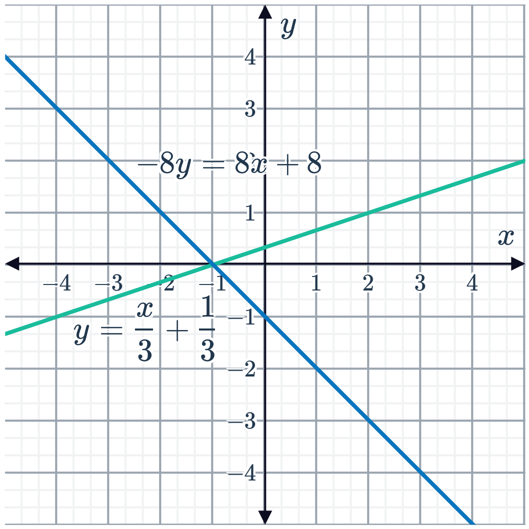
b $x=1, y=5$

6

a $(-1,0),\left(0,\dfrac{1}{3}\right)$

b $(0,-1),(-1,0)$

c



d $x=-1, y=0$

7

a $x=6, y=4$

b $x=0, y=-2$

c $x=4, y=0$

8

a $x=8, y=3$

b $x=\dfrac{20}{27}, y=\dfrac{62}{27}$

c $x=-20, y=15$

d $x=2, y=\dfrac{5}{3}$

e $p=\dfrac{2}{5}, q=\dfrac{3}{4}$

f $p=\dfrac{3}{5}, q=\dfrac{4}{5}$

9

a $x=4, y=2$

b Yes

c Yes

10 $x+y=9, y-x=1$

11

a $x=13$

b $x=9$

12

a 4

b $x=-3, y=-9$

c No

13

a $x = 1, y = 9$

b $y = -3x + 12$

c $3y = 3x + 24$

d $x = 1, y = 9$

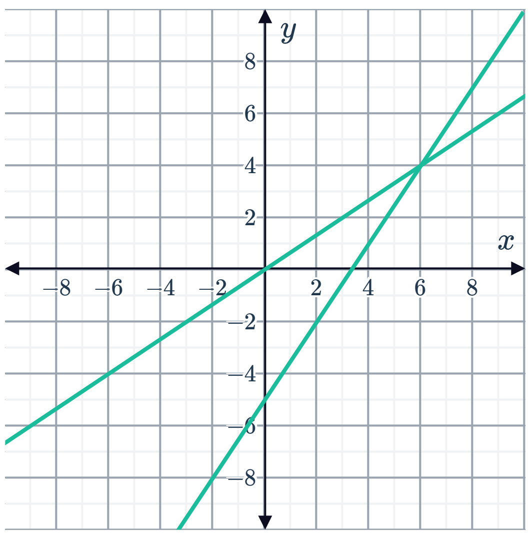
e Yes, Equations 3 and 4 are the result of applying identical operations to both the left hand side and the right hand side of each of the original equations.

14

a $x = 6$

b Equation 1 and Equation 4

c



d $x = 6$

15

a

i

x	1	3	5	7	9
y	58	64	70	76	82

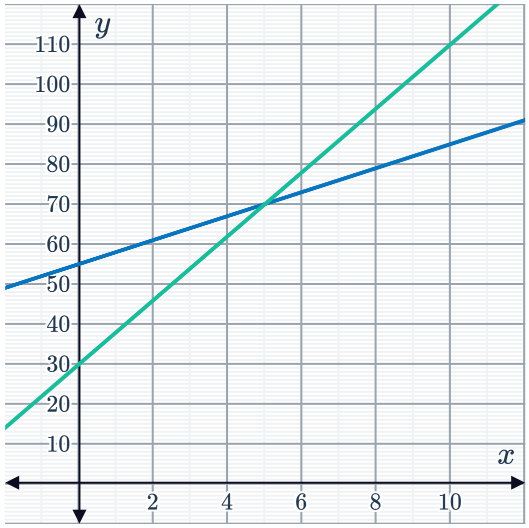
ii

x	1	3	5	7	9
y	38	54	70	86	102

b

i No ii No iii Yes iv No

c



5 minutes

e TeleNow

16

a No

b Yes

c No

d No

Additional questions

17

a $(0,4)$

b $(4,6)$

c $(8,9)$

d $(-3,4)$

e $(7,2)$

f $(9,6)$

18

a $(5,-1)$

b $(-2,6)$

c No solutions

d $(7,-6)$

19

a $(-3.125, 7.9375)$

b $\left(33, -\frac{100}{19}\right)$

c $\left(\frac{13}{3}, 36\right)$

d $\left(\frac{15147}{37}, \frac{347}{37}\right)$